Indian Institute of Technology Bhilai

Course Project Report

Network Science

Network Analysis of Escalation, Bridge Actors, and Cross-Theater Conflict Structure

in the US–Israel–Iran Conflict System

(2015–Latest Available Data)

Submitted by

Om Raj Singh (M25DS007)
Rohan Sinha (M25DS008)
Sarvesh Badoni (M25DS011)
Vedant Tawri (M25DS016)

Indian Institute of Technology Bhilai

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Contents

1 Abstract **3**

2 Introduction **3**

3 Why This Project Matters **3**

4 Research Objective and Research Questions **4**

4.1 Objective 4

4.2 Research Questions 4

5 Data Sources, Scope, and Phase Design **4**

5.1 Source Datasets 4

5.2 Conflict Scope 5

5.3 Phase Design 5

6 Data Preparation Pipeline **5**

6.1 Harmonization 5

6.2 Strict Filtering 5

6.3 Phase Labeling 6

6.4 Why These Steps Matter 6

7 Methodology **7**

7.1 Actor–Actor Interaction Network 7

7.2 Event–Actor Bipartite Representation 7

7.3 Network Metrics 7

7.4 Community Detection and Bridge Scoring 8

8 Descriptive Profile of the Final Dataset **8**

9 Answer to Research Question 1: How Does the Network Evolve Across Phases? **10**

10 Answer to Research Question 2: Which Actors Act as Bridges Across Theaters? **12**

11 Answer to Research Question 3: Are the Most Important Actors Simply the Most Active Ones? **13**

12 Answer to Research Question 4: Is the Recent Phase Structurally Different? 14

13 Answer to Research Question 5: What Changes in Country and Event Composition Accompany the Later Phase? 15

14 Overall Network Structure and Community Interpretation 16

15 Phase-specific Network Interpretation 19

15.1 Phase 1: Transitional Shadow Phase 19

15.2 Phase 2: Escalatory Hybrid Phase 19

15.3 Phase 3: Current Escalation Phase 20

16 Fatality Context 23

17 Interpretation of Findings 23

18 Limitations 24

19 Conclusion 24

1 Abstract

This report presents a network-science study of the US–Israel–Iran conflict system using event data from 2015 to the latest available date in the assembled dataset. The analysis begins with a harmonized ACLED–GED event base, applies a strict geographic filter for the core conflict theater, and divides the final dataset into three phases: 2015–2018, 2019–2021, and 2022 onward. On this foundation, we construct actor–actor networks, event–actor bipartite representations, and phase-wise structural summaries to study escalation, fragmentation, community structure, and bridge actors across the regional conflict system.

The report is motivated by a simple but important question: whether the recent phase of the conflict is only more intense, or whether it is also structurally different from earlier periods. The findings support the second interpretation. Across phases, the actor network grows in size, the number of communities rises sharply, and bridge positions become more important for understanding how otherwise separate theaters remain connected. The analysis also shows that structurally important actors are not always identical to the most active actors by volume. In this sense, the main contribution of the project is to use standard network-science tools to reveal non-obvious cross-theater relations and actor roles that are not visible from event counts alone.

2 Introduction

Regional conflicts are rarely confined to a single front, a single state, or a single dyadic rivalry. They evolve through repeated interactions among state forces, militias, proxy organizations, coalitions, and civilian populations across multiple theaters. In such settings, the main analytical challenge is not only to measure how much violence is occurring, but also to understand how the wider conflict system is organized.

The US–Israel–Iran conflict system is an appropriate case for this kind of analysis. It involves direct and indirect confrontation across Israel/Palestine, Lebanon, Syria, Iraq, Yemen, Iran, and US-linked developments. Public discussion often treats these theaters separately or focuses on major headline events, but many of the important dynamics of escalation emerge from the way actors and theaters connect to one another over time.

This project approaches the problem through network science. Instead of studying the conflict only through yearly counts or country-wise summaries, it represents the system as a set of interacting actors and evolving structural relations. This allows the analysis to ask questions that are harder to answer using ordinary descriptive methods: which actors connect otherwise separate arenas, how conflict communities reorganize across phases, and whether the current escalation phase is merely larger or structurally distinct.

3 Why This Project Matters

The project was designed to remain within the scope of a course-level network-science study while still addressing a meaningful substantive problem. A simpler project might have ended with trend charts and centrality rankings. A much more advanced project might have moved into causal modeling, multiplex networks, or formal temporal sequence models. This work takes

a middle path.

Its novelty does not lie in proposing a new mathematical method. The novelty lies in how standard network-science methods are used to answer a sharper substantive question. The project treats the US–Israel–Iran conflict not as a set of isolated national event streams, but as a connected multi-theater system. It emphasizes bridge actors, community structure, and phase-wise structural change. In doing so, it attempts to show something more informative than simple escalation by volume: namely, how the organization of the conflict system itself changes over time.

4 Research Objective and Research Questions

4.1 Objective

The main objective of the project is to analyze the structural evolution of the US–Israel–Iran conflict system from 2015 to the latest available data, and to identify the actors, communities, bridge relations, and broad escalation-linked structural changes that shape the conflict system in the contemporary period.

4.2 Research Questions

The report is organized around the following research questions:

1. How has the actor interaction network evolved across the phases 2015–2018, 2019–2021, and 2022 onward?
2. Which actors function as bridges across otherwise separate theaters such as Israel/Palestine, Lebanon, Syria, Iraq, Yemen, and Iran-linked developments?
3. Are the most structurally important actors simply the most active actors, or are they the ones that connect different communities and theaters?
4. Does the recent phase differ from earlier phases only in size, or also in community structure, fragmentation, and cross-theater connectivity?
5. What broad changes in country participation and event-type composition accompany the transition into the recent escalation phase?

These questions are important because they move the analysis beyond descriptive intensity. They are also directly answerable using the network pipeline developed in the project.

5 Data Sources, Scope, and Phase Design

5.1 Source Datasets

The starting point of the project is a harmonized event dataset built from ACLED and multiple GED releases. ACLED provides the main event base for the project, while GED is integrated

through a schema-harmonization step so that the combined data can be used in a common pipeline. Each row in the harmonized output retains source metadata, source file identity, a synthetic flag, and harmonization-confidence information.

5.2 Conflict Scope

After harmonization, the dataset is filtered to a strict geographic conflict scope consisting of the following countries:

Israel, Palestine, Lebanon, Syria, Iraq, Yemen, Iran, and the United States.

This scope was chosen to keep the analysis centered on the conflict system that the project intends to study. It avoids turning the report into a general Middle East event inventory while still preserving the main theaters through which US–Israel–Iran regional dynamics unfold.

5.3 Phase Design

The final filtered dataset is divided into three analytical phases:

- **Phase 1 (2015–2018):** Transitional Shadow Phase
- **Phase 2 (2019–2021):** Escalatory Hybrid Phase
- **Phase 3 (2022–2026):** Current Escalation Phase

This phase design supports meaningful comparison without creating very short and unstable sub-periods. It also aligns well with the project goal of understanding whether the recent phase is structurally different from earlier periods.

6 Data Preparation Pipeline

6.1 Harmonization

The first stage of the pipeline creates a unified conflict-event dataset from ACLED and GED inputs. ACLED rows are treated as primary source records, while GED rows are mapped into a compatible schema with standardized date, actor, location, and event fields. Duplicate GED event identifiers across releases are removed by retaining the latest version.

6.2 Strict Filtering

The second stage applies a country-only filter to retain only rows belonging to the eight-country conflict scope. This narrows the harmonized dataset into a theater-specific corpus suitable for regional structural analysis.

6.3 Phase Labeling

The filtered dataset is then labeled phase-wise according to event year. Additional fields for phase and phase label are added, and the segmented output becomes the main input for network construction.

6.4 Why These Steps Matter

These preprocessing stages are not merely technical preparation. They are part of the substantive logic of the project. The harmonization step allows multiple sources to be used consistently; the strict filter defines the actual conflict system under study; and the phase labels enable the comparison required to answer the research questions.

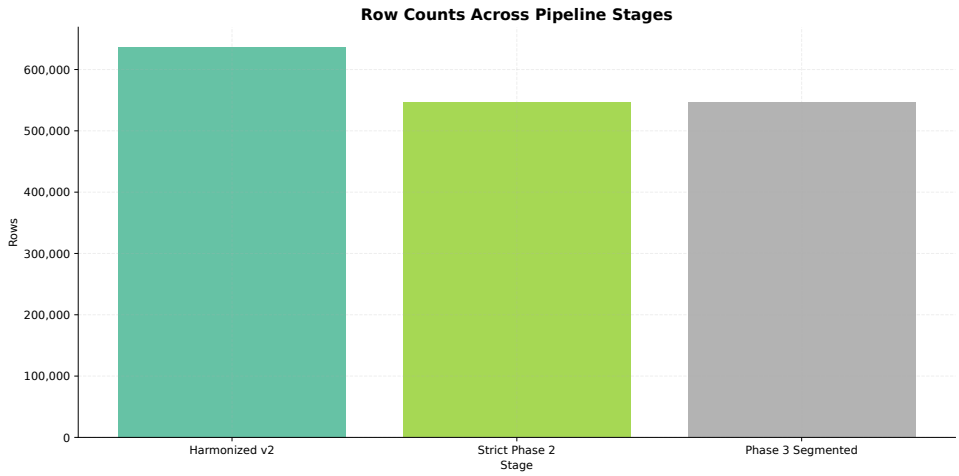


Figure 1: Row counts across the major pipeline stages. The figure shows how the harmonized event base is narrowed into the final segmented analysis dataset.

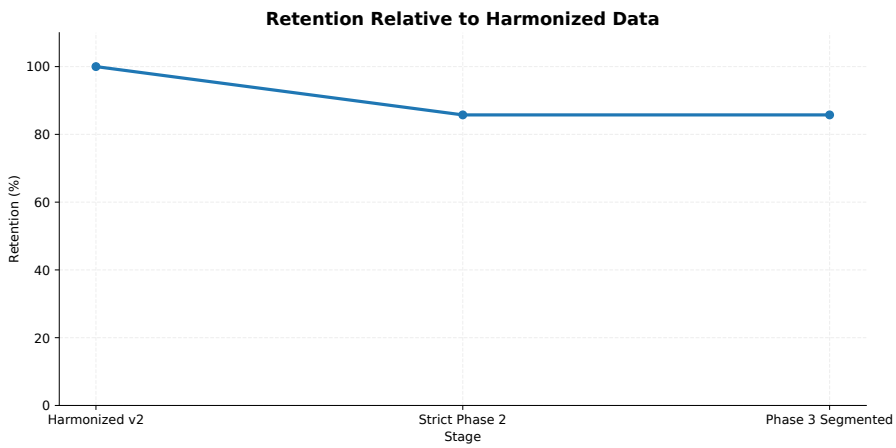


Figure 2: Retention relative to the harmonized dataset across downstream stages. The filtered and segmented outputs preserve the main conflict-relevant event base while enforcing a narrower scope.

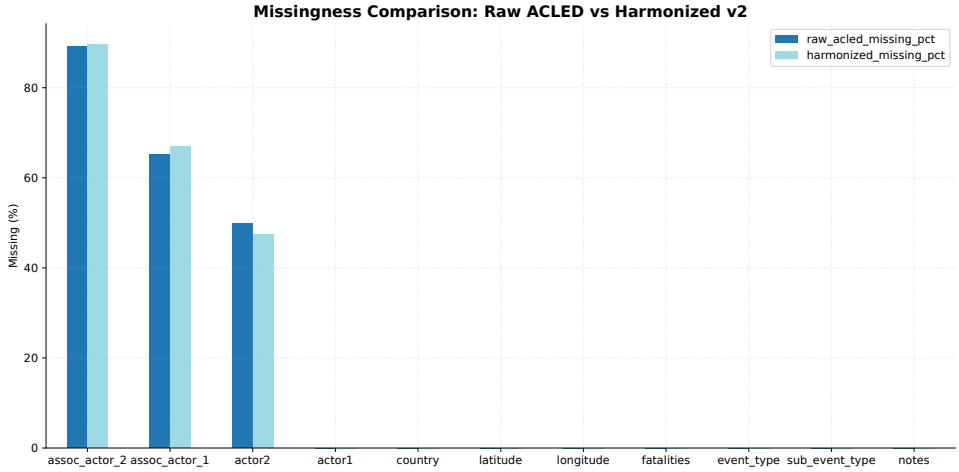


Figure 3: Comparison of missingness in selected fields between raw ACLED input and the harmonized dataset. Harmonization improves schema consistency and downstream usability.

7 Methodology

7.1 Actor–Actor Interaction Network

The main analytical representation in the report is an undirected weighted actor–actor network. For each event where both `actor1` and `actor2` are present, an actor pair is created and treated symmetrically through canonical ordering. Edge weight corresponds to the number of unique events in which the actor pair appears together.

This network is central to the report because it allows direct investigation of the first four research questions: phase-wise evolution, bridge actors, structural importance, and differences between earlier and later phases.

7.2 Event–Actor Bipartite Representation

The project also constructs a bipartite event–actor edge list by linking each event to its participating actors. Although this report focuses mainly on the actor network, the bipartite structure preserves event-level participation and provides a more grounded representation of actor involvement.

7.3 Network Metrics

For the overall network and for each phase-specific network, the report computes:

- degree and weighted degree,
- degree centrality,
- betweenness centrality,
- eigenvector centrality,

- clustering coefficient,
- density,
- connected components,
- largest component size.

These metrics are sufficient to distinguish activity from structural position and to compare how network organization changes across phases.

7.4 Community Detection and Bridge Scoring

The report uses the Louvain method for community detection on weighted actor networks. This produces community assignments and modularity scores for the overall graph and each phase-specific graph. Community detection is important because it allows the report to study bloc structure and fragmentation rather than only central nodes.

Bridge actors are identified through a bridge score that combines betweenness centrality, degree centrality, and normalized weighted degree. This is one of the most important methodological contributions of the project, because it operationalizes a distinction between actors that are merely active and actors that are structurally connective.

8 Descriptive Profile of the Final Dataset

Before moving to network findings, it is useful to establish the broad composition of the final dataset. After strict filtering, the corpus remains overwhelmingly ACLED-based, with GED contributing only a small fraction. This means that the report is primarily interpreting an ACLED-centered event structure, even though the harmonization step broadens the initial event base.

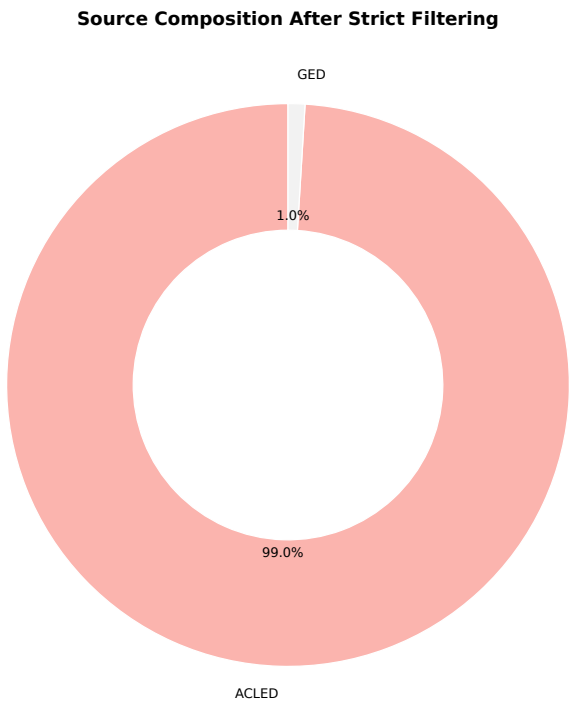


Figure 4: Source composition after strict filtering. The final dataset is overwhelmingly ACLED-based, with GED contributing a small residual share.

The country composition of the filtered dataset is uneven. Syria and Palestine account for a large share of the events, but this does not mean that other theaters are analytically unimportant. One of the arguments of this report is precisely that some theaters matter because they help connect the broader system, not only because they contribute the largest event counts.

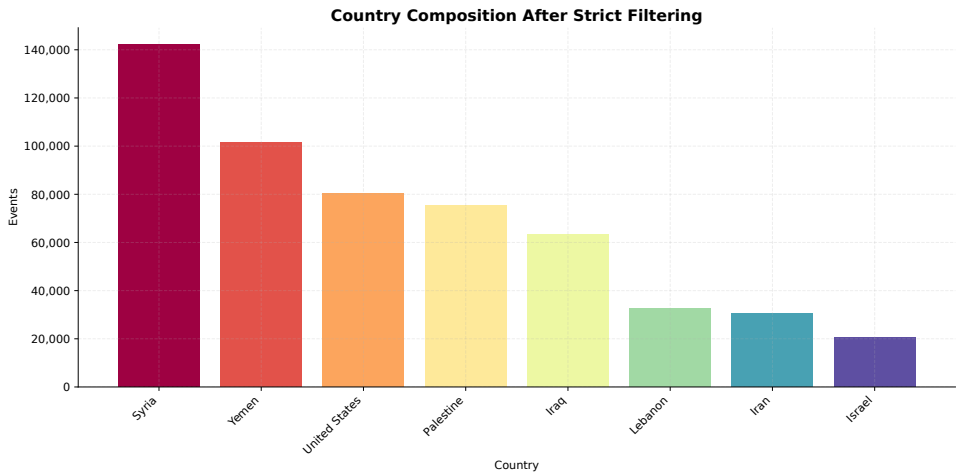


Figure 5: Country composition after strict filtering. Syria and Palestine dominate event volume, while the remaining theaters contribute to the larger multi-arena structure of the conflict system.

Phase distribution is also important for interpretation. The most recent phase accounts for the largest share of the final segmented dataset, which makes it both substantively central and statistically suitable for comparison.

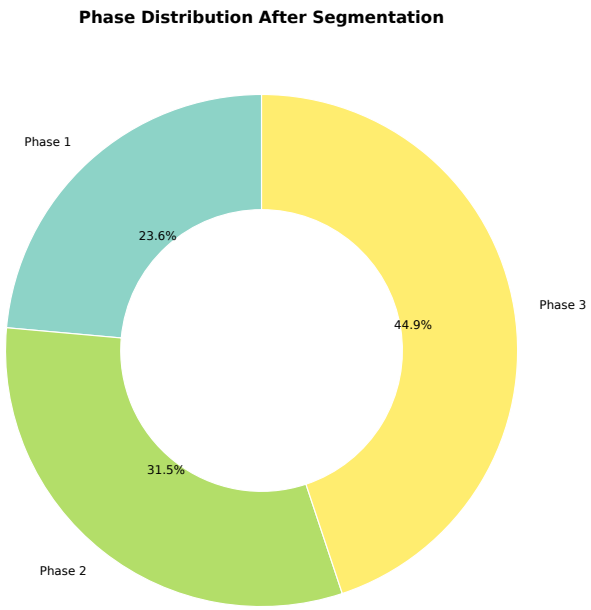


Figure 6: Distribution of events across the three analytical phases. The most recent phase contains the largest share of the segmented dataset.

9 Answer to Research Question 1: How Does the Network Evolve Across Phases?

The first research question asks how the actor interaction network evolves across the three phases. The clearest answer is that the network expands substantially over time. Nodes, edges, communities, and connected components all increase from Phase 1 to Phase 3. This means that the later conflict environment is not simply a repetition of earlier structures with more events added. It is a broader and more differentiated interaction system.

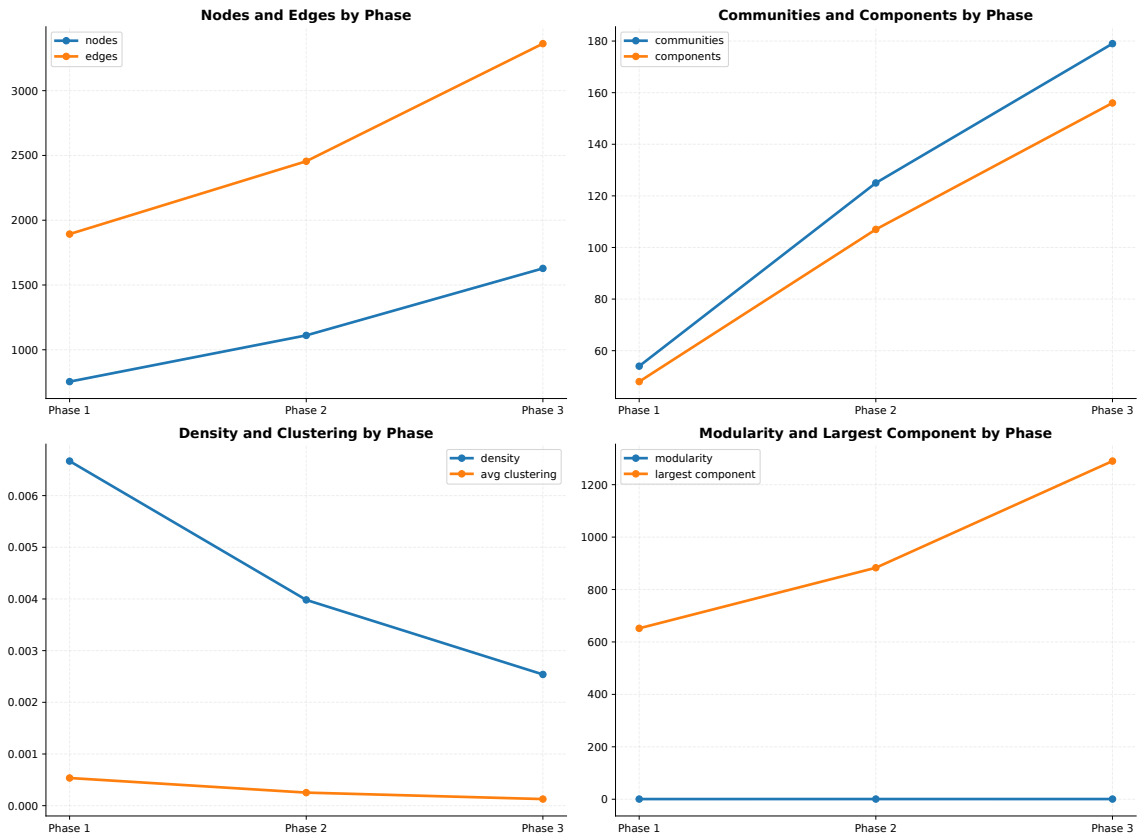


Figure 7: Phase-wise structural dashboard. The figure shows growth in nodes, edges, communities, and components across phases, alongside declining density and clustering as the network expands.

The dashboard also reveals an important nuance. As the network grows, density declines. This is not a contradiction. In empirical networks, density often falls when the number of possible actor pairs grows faster than the number of realized ties. The decline in density therefore suggests that the later conflict system becomes broader and more segmented rather than uniformly tighter.

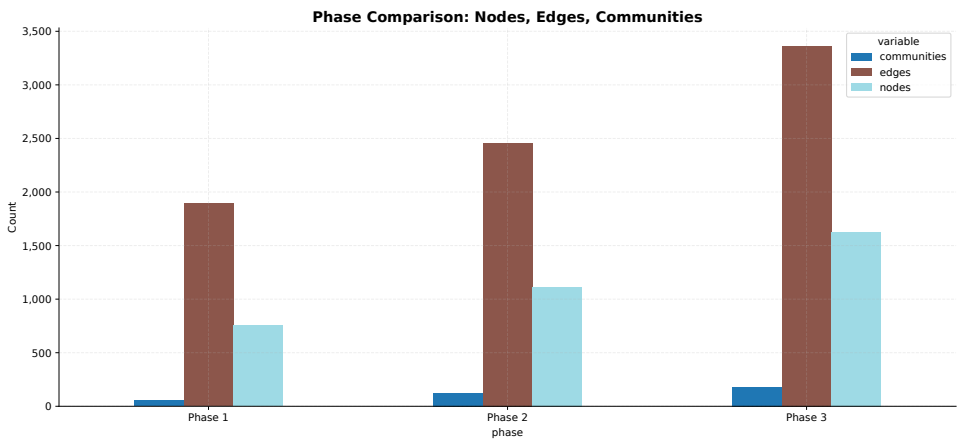


Figure 8: Comparison of nodes, edges, and communities across phases. Structural expansion is visible in each successive phase.

This is one of the core findings of the report. The later phase is not only larger by volume; it is structurally different in the sense that it contains more actors, more ties, and more differentiated

clustering. That is one of the main novelties of the analysis.

10 Answer to Research Question 2: Which Actors Act as Bridges Across Theaters?

The second research question asks which actors act as bridges across otherwise separate conflict theaters. This is where the report moves beyond standard event-count analysis. Instead of looking only at actor frequency, it identifies actors that occupy structurally connective positions between communities and subregions of the graph.

Across the overall network, several actors emerge as important bridges. These actors are not necessarily important only because of event volume. Their importance lies in the fact that they connect parts of the conflict system that would otherwise be more weakly linked. In a multi-theater conflict, such actors are especially important because they help preserve regional connectivity.

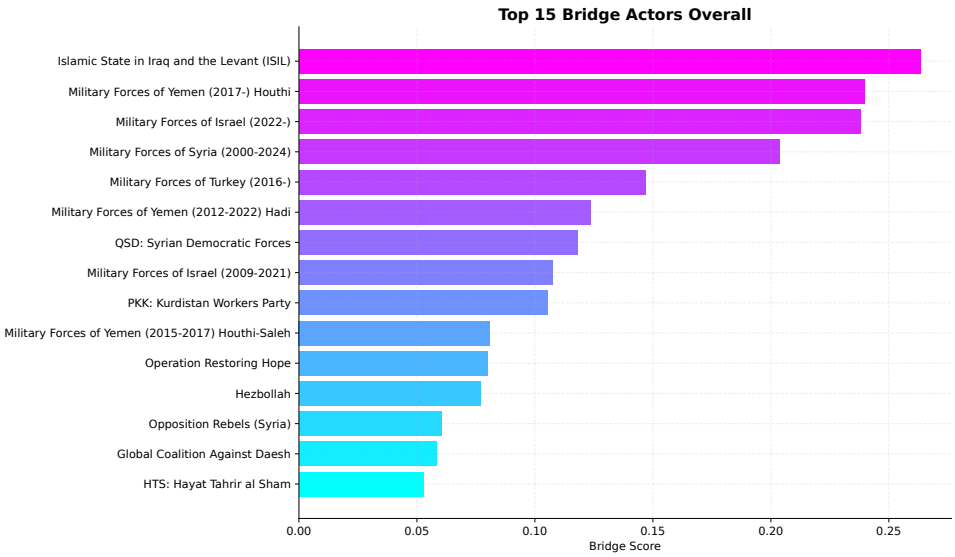
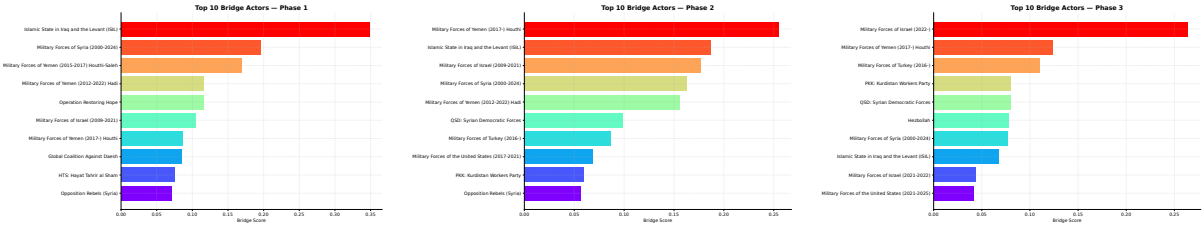


Figure 9: Top 15 bridge actors in the overall network. These actors are structurally important connectors across communities and theaters.

The phase-wise bridge plots add a second layer of interpretation. The identity of important bridge actors changes across phases. Phase 1 is more strongly shaped by Syria- and Yemen-linked actors; Phase 2 brings in additional Turkish, Kurdish, and US-linked connectors; and Phase 3 makes Israel-, Hezbollah-, and broader regional connectors more visible. This suggests that the conflict system is not only expanding, but reorganizing.



(a) Phase 1 **(b)** Phase 2 **(c)** Phase 3

Figure 10: Top bridge actors by phase. The identity of important connectors shifts across phases, indicating structural reorganization rather than simple persistence.

This is one of the strongest parts of the report. The novelty here is not mathematical complexity, but the substantive insight that actors linking theaters may matter more for understanding regional escalation than actors identified only through volume-based rankings.

11 Answer to Research Question 3: Are the Most Important Actors Simply the Most Active Ones?

The third research question asks whether structurally important actors are simply the most active actors. The results suggest that the answer is only partially yes. Weighted degree identifies actors with many or strong connections, but betweenness highlights actors that connect different parts of the network. These are related forms of importance, but they are not identical.

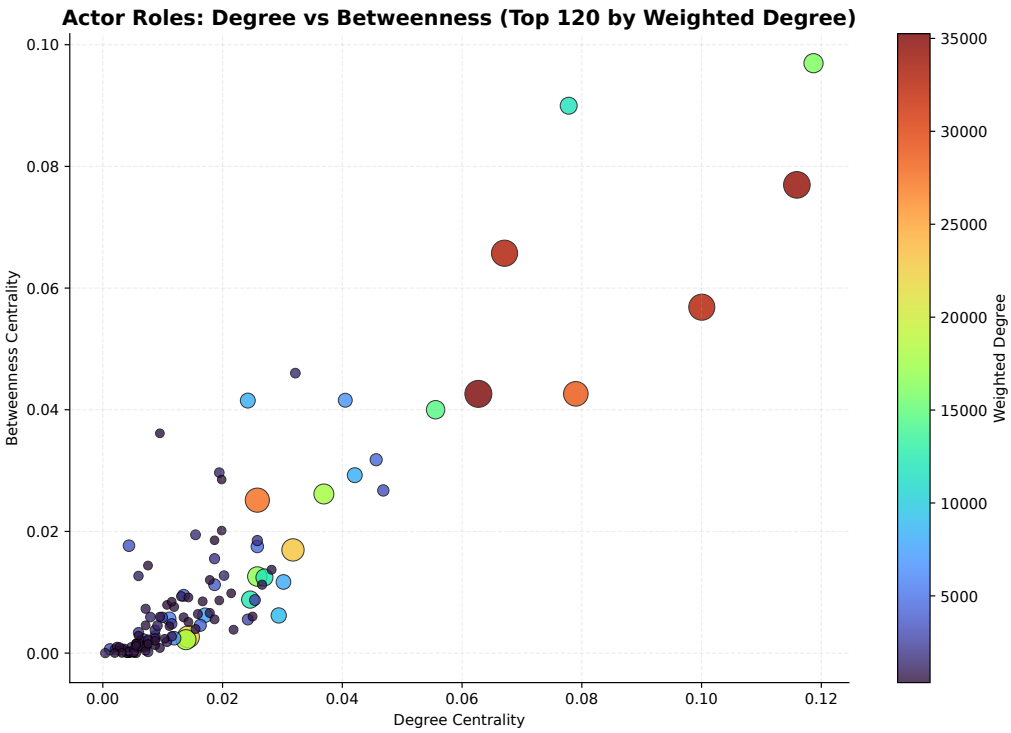
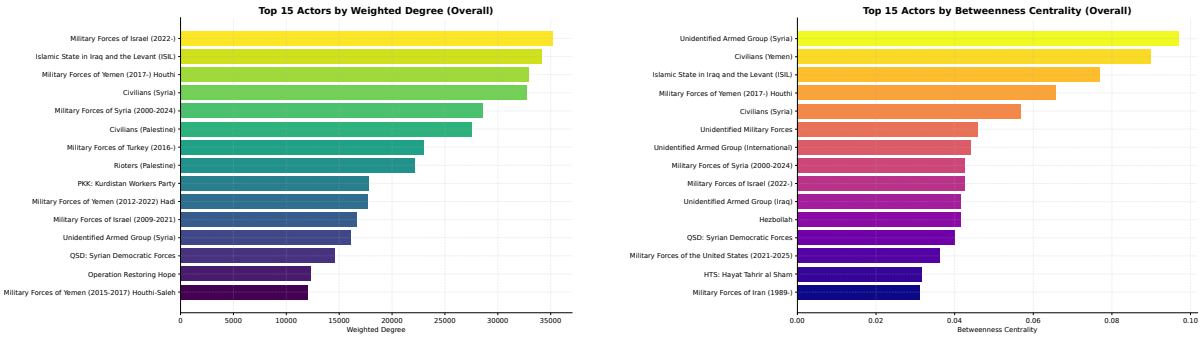


Figure 11: Actor roles by degree and betweenness. Not all highly connected actors are equally important as bridges.

The spread visible in Figure 11 is analytically important. Some actors combine high degree and high betweenness, which means they are both active and structurally central. Others have high

degree but lower betweenness, indicating that they are embedded inside dense local subgraphs rather than linking distinct communities. This directly answers the research question: structural importance cannot be reduced to raw activity.



(a) Top actors by weighted degree

(b) Top actors by betweenness centrality

Figure 12: Overall actor rankings using two different notions of structural importance.

This distinction is one of the non-trivial findings of the project. It demonstrates why a network-science perspective is useful: simple activity rankings alone would miss the difference between actors that dominate locally and actors that connect the wider regional system.

12 Answer to Research Question 4: Is the Recent Phase Structurally Different?

The fourth research question asks whether the recent phase differs from earlier phases only in size, or also in structural organization. The evidence strongly supports the second interpretation.

The most recent phase contains more actors, more edges, more communities, and a larger component structure than the earlier phases. At the same time, the network becomes less dense and less tightly clustered. This combination suggests a broader and more fragmented system rather than a uniformly intensified one. In other words, the recent phase appears structurally wider, not merely heavier.

The community results support the same conclusion. Community counts rise substantially from Phase 1 to Phase 3, while modularity remains high. This means that the later phase is not a random proliferation of ties. It is a larger system that still preserves interpretable internal organization.

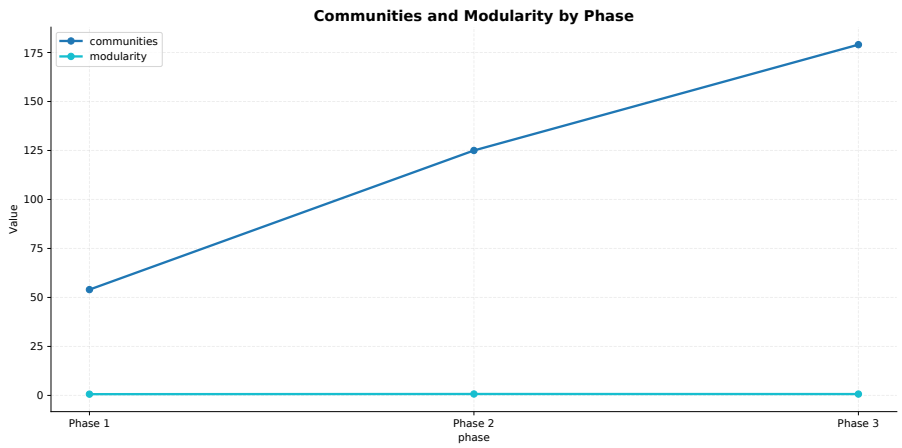


Figure 13: Communities and modularity by phase. Community counts increase strongly over time while modularity remains high.

This is perhaps the central novelty of the report. The project shows that the recent escalation phase is not only more intense by event volume, but structurally different in terms of fragmentation, clustering, and cross-theater organization.

13 Answer to Research Question 5: What Changes in Country and Event Composition Accompany the Later Phase?

The fifth research question focuses on broader composition shifts. The country-by-phase matrix shows that some theaters remain consistently large, while others become more prominent later in the timeline. Syria remains central throughout, while Palestine, Iraq, Yemen, and the United States become especially prominent in the most recent phase.

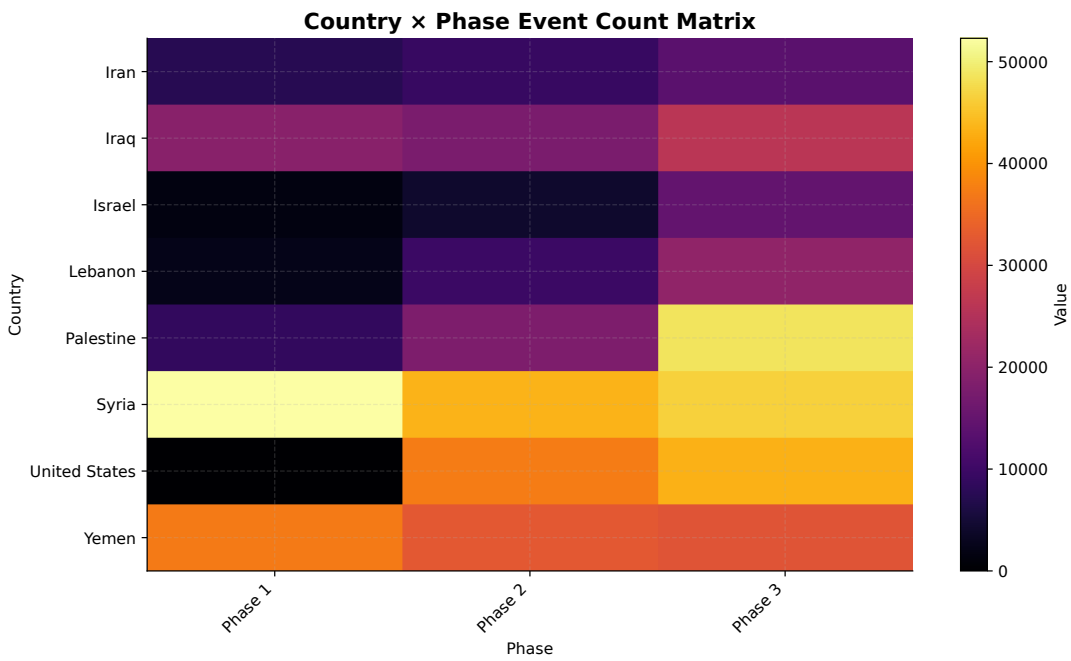


Figure 14: Country × phase event-count matrix. The later phase shows especially strong event concentration in Palestine, Syria, Iraq, Yemen, and the United States.

Event-type composition also broadens in the later phase. Strategic developments, riots, protests, battles, remote violence, and violence against civilians all contribute substantially. This suggests that the later conflict environment is not dominated by a single form of contention, but by overlapping military, strategic, and societal modes of escalation.

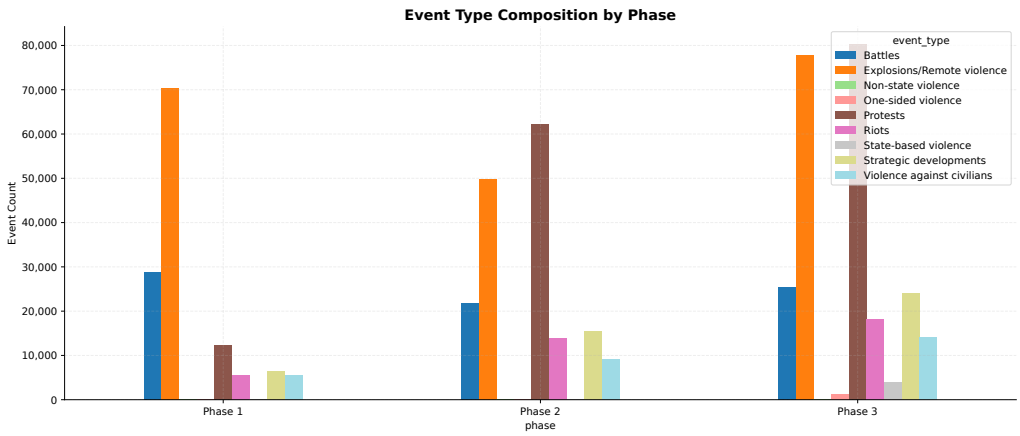


Figure 15: Event-type composition by phase. The most recent phase combines increased volume with broader composition across event categories.

These composition shifts strengthen the structural interpretation developed in the earlier sections. The recent phase is wider in both geographic and event-form terms, which is consistent with the network evidence for broader multi-theater conflict organization.

14 Overall Network Structure and Community Interpretation

The overall actor network reveals a multi-cluster structure rather than a single integrated block. The graph contains dense subregions, peripheral attachments, and visible long-range links. This matters because it suggests that the conflict system is regionally connected without being homogeneous.

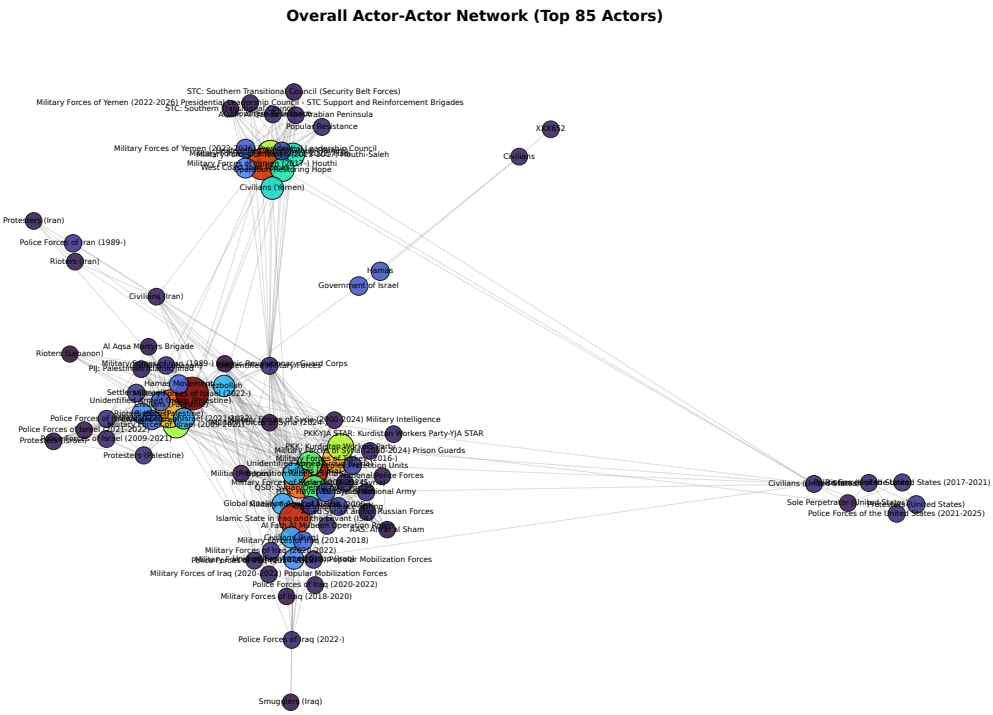


Figure 16: Overall actor–actor network for the top 85 actors. The layout shows a dense core with multiple attached substructures and visible long-range connections.

The community-colored version makes this even clearer. Distinct communities are visible, but they are not isolated in a strict sense. Some actors and ties keep the broader graph connected across otherwise separate clusters.

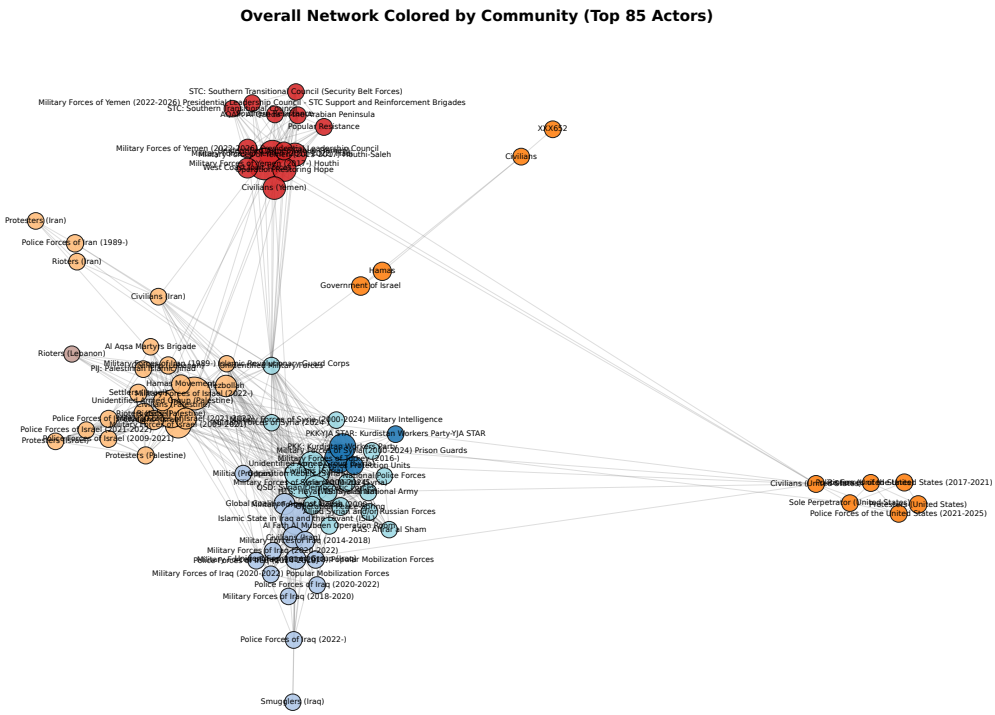


Figure 17: Overall network colored by community. Distinct actor blocs are visible, but they remain part of a wider connected system.

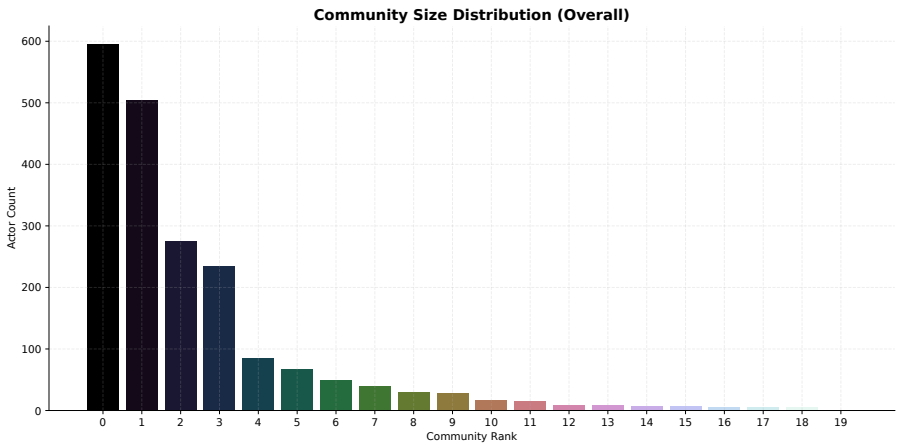


Figure 18: Overall community-size distribution. A small number of large communities coexist with many smaller ones, indicating hierarchical fragmentation.

These figures are important to the novelty of the report because they visually support the main claim: the conflict should be understood as a connected but internally differentiated system rather than as a uniform block of events.

15 Phase-specific Network Interpretation

15.1 Phase 1: Transitional Shadow Phase

Phase 1 is smaller and relatively compact. Major conflict theaters are already visible, especially in Syria- and Yemen-linked portions of the network, but the total number of actors, edges, and communities remains limited. This phase looks like a structurally narrower regional system.

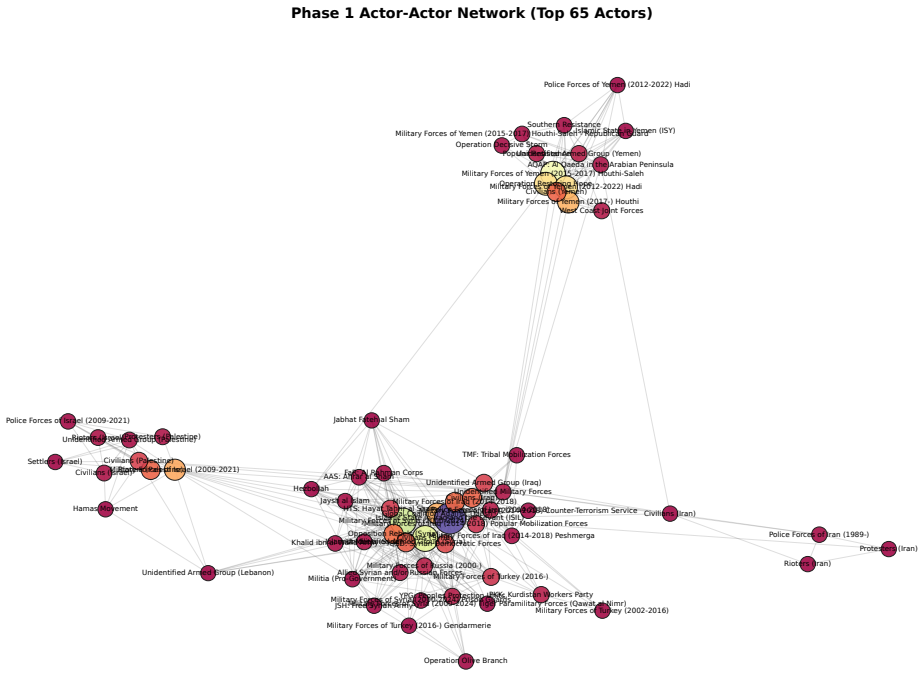


Figure 19: Actor–actor network in Phase 1. The structure is relatively compact, with major early theaters already visible.

15.2 Phase 2: Escalatory Hybrid Phase

Phase 2 is noticeably more expanded. Additional clusters and linked substructures become visible, and the graph begins to show stronger regional spread. This phase is analytically important because it appears to mark a structural transition rather than a simple intermediate period.

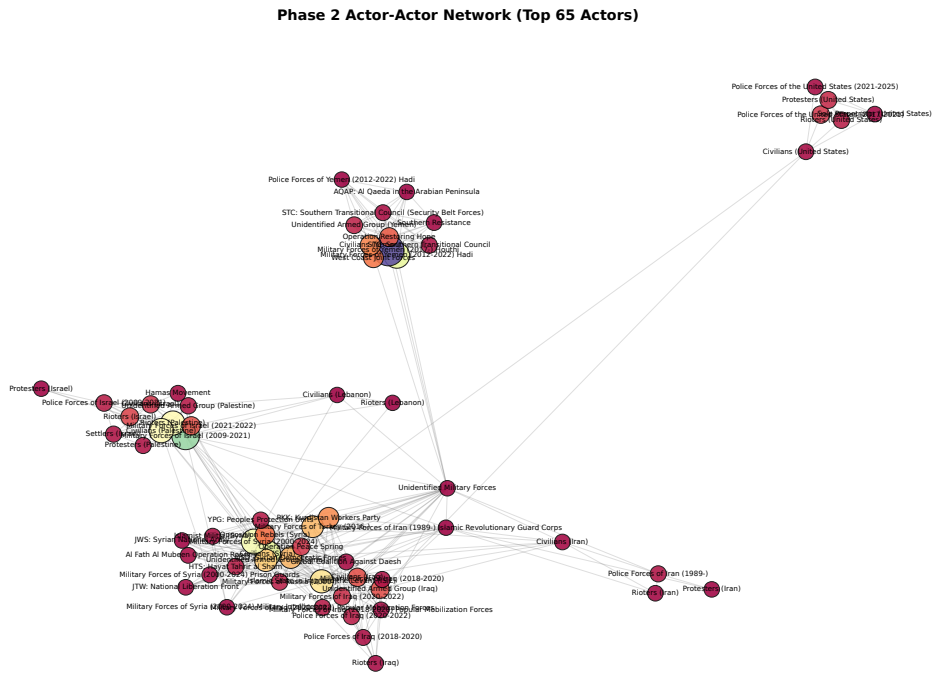
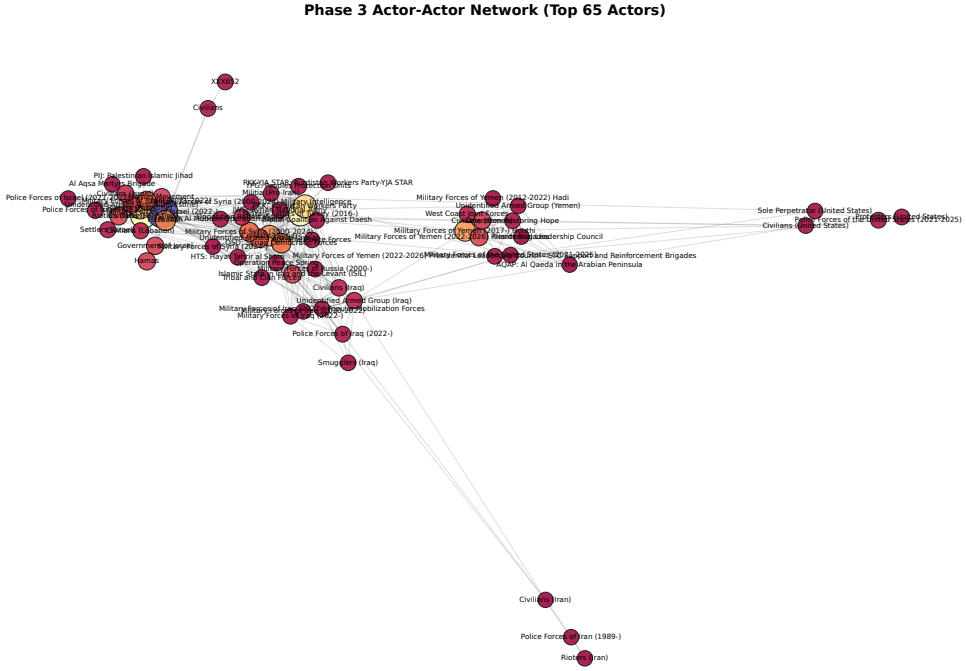
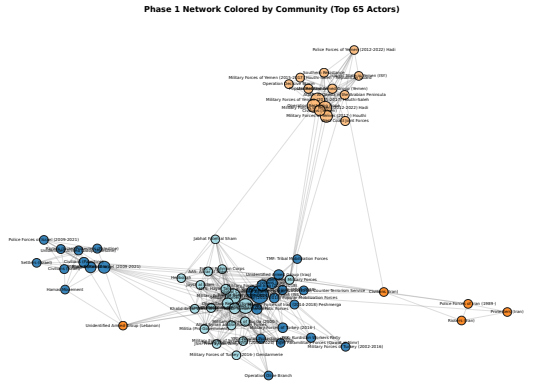


Figure 20: Actor–actor network in Phase 2. Compared with Phase 1, the graph is more expanded and reveals additional linked substructures.

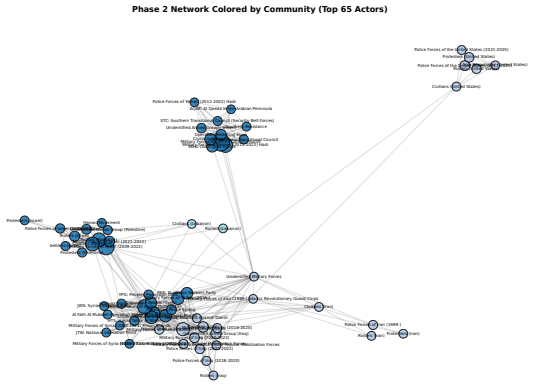
15.3 Phase 3: Current Escalation Phase

Phase 3 is the largest and most differentiated network in the report. Multiple theaters are strongly visible, community structure broadens, and bridge actors become more salient. The graph is visibly more fragmented and more cross-theater in form than the earlier phases.

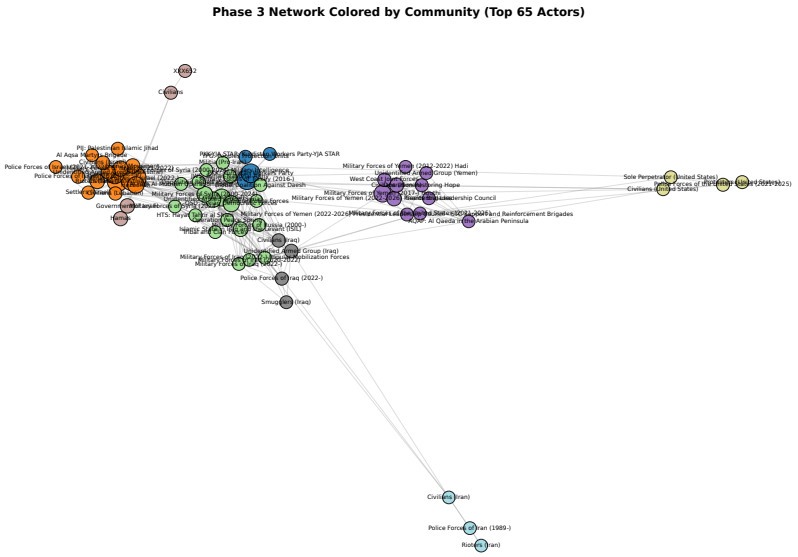

Phase 3 Actor-Actor Network (Top 65 Actors)
Civilians (Iran)
Police Forces of Iran (1989-)
Riot (Iran)
Smuggler (Iraq)
Police Forces of Iraq (2022-)
Civilians (Iraq)
Unidentified Armed Group (Iraq)
Sole Perpetrator (United States)
Civilians (United States)
Military Forces of Yemen (2012-2022) Hadi
Unidentified Armed Group (Yemen)
Leadership Council
Reinforcement Brigades
AQAP: Al Qaeda in the Arabian Peninsula
Military Forces of Yemen (2022-2026)
Military Forces of Yemen (2017-2022) Saleh
Military Forces of Yemen (2022-2026)
Military Forces of Syria (2024-)
Islamic State (ISIL)
Hamas
Settler (Israel)
Government of Israel
Al Aqsa Martyrs' Brigade
Police Forces of Israel
PIJ: Palestinian Islamic Jihad
Military Forces of Russia (2000-)
Military Forces of Syria (2024-)
Militia (Iraq)
Militia (Syria)
Militia (Yemen)
Rioter (United States)
Protestor (United States)
Protest (United States)
Civilians (Syria)
Civilians (Yemen)
Civilians (Israel)
Civilians (Palestine)
Civilians (Russia)
Civilians (Iraq)
Civilians (Lebanon)
Civilians (Turkey)
Civilians (Egypt)
Civilians (Jordan)
Civilians (Saudi Arabia)
Civilians (Kuwait)
Civilians (Qatar)
Civilians (Bahrain)
Civilians (Oman)
Civilians (Afghanistan)
Civilians (Pakistan)
Civilians (India)
Civilians (China)
Civilians (Japan)
Civilians (Germany)
Civilians (France)
Civilians (Italy)
Civilians (Spain)
Civilians (Portugal)
Civilians (Greece



(a) Phase 1

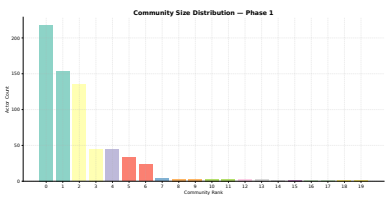


(b) Phase 2

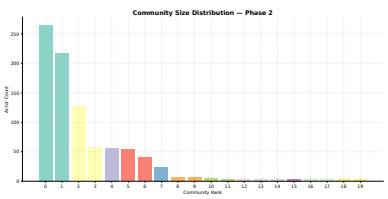


(c) Phase 3

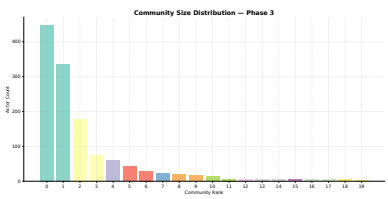
Figure 22: Community-colored phase-wise networks. Later phases exhibit larger, more differentiated, and more visibly cross-linked community structure.



(a) Phase 1



(b) Phase 2



(c) Phase 3

Figure 23: Community-size distributions by phase. The later phase contains a broader distribution of community structure.

These phase-wise interpretations are important because they provide a direct answer to the overall research agenda of the project: the contemporary conflict phase is structurally wider, more differentiated, and more dependent on cross-theater connectors than the earlier phases.

16 Fatality Context

The report is primarily structural rather than severity-driven, but fatality summaries still provide useful context. Fatality totals differ across phases, and the event distribution remains highly skewed. This is one reason the report does not reduce the conflict system to severity counts alone.

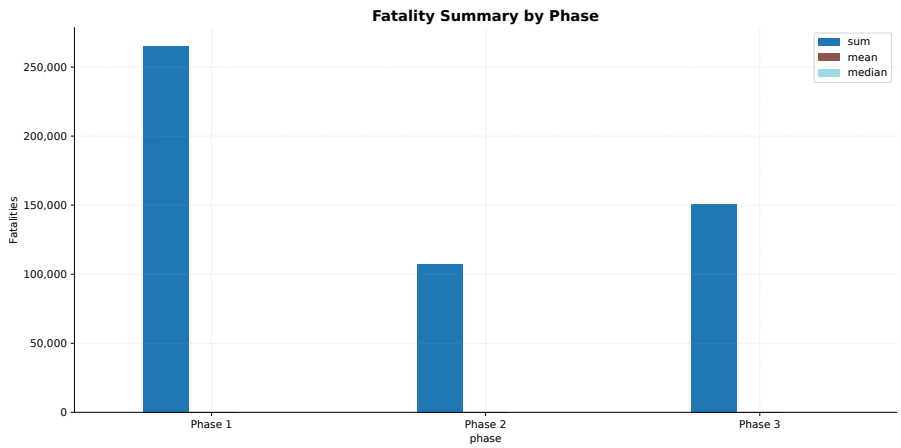


Figure 24: Fatality summary by phase. The phases differ not only in network size but also in severity context.

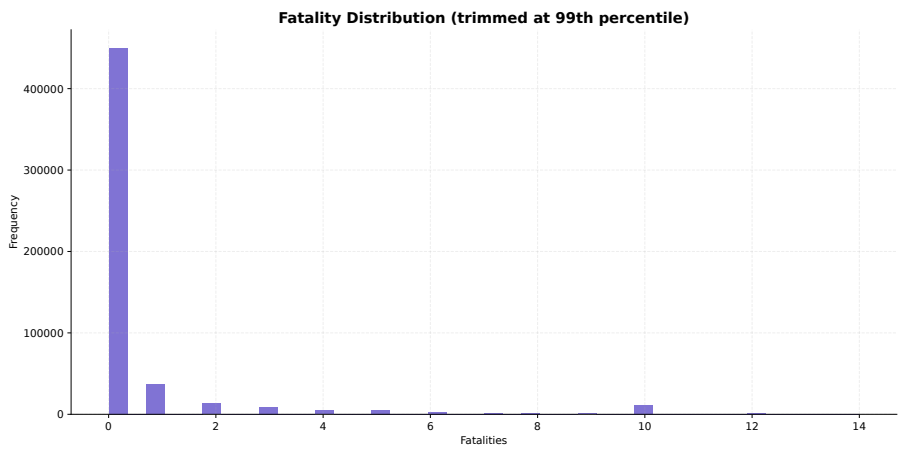


Figure 25: Fatality distribution after trimming the extreme tail. The underlying event distribution remains highly skewed.

17 Interpretation of Findings

The report set out to examine whether the recent phase of the US–Israel–Iran conflict system is simply more intense or whether it is structurally different. The findings support the second interpretation.

First, the actor interaction network clearly expands across phases. More actors enter the system, more ties appear, and the number of communities rises strongly. This already indicates that the later phase is broader in structural scope than the earlier phases.

Second, the bridge analysis shows that regional connectivity depends on more than highly active actors. Some actors matter because they help connect distinct communities and theaters. This is one of the most important findings of the project, because it provides a structural explanation for cross-theater conflict organization that ordinary event counts cannot easily show.

Third, the community results show that expansion does not eliminate structure. Even as the network grows, modularity remains high and meaningful clusters persist. This suggests that the conflict system is widening while still retaining bloc-like organization.

Fourth, country and event-type shifts reinforce the same conclusion. The later phase is broader not only in actor-network terms, but also in the spread of theaters and forms of conflict activity represented in the data.

Taken together, these results explain the novelty of the project. The report does not claim novelty by inventing a new method. Its novelty lies in using standard network tools to reveal that the current phase is structurally different from earlier ones, and that bridge actors and cross-theater relations are central to that difference.

18 Limitations

The findings should still be interpreted within the limits of the data and modeling choices.

First, the final filtered dataset remains heavily ACLED-dominated, even though GED broadens the initial event base. The resulting network structure therefore still reflects ACLED coverage and actor coding to a large extent.

Second, the actor–actor network is built from co-occurrence within recorded events. This is a useful and interpretable structural approximation, but it is not a direct representation of command, alliance, or intent.

Third, community detection and bridge scoring are modeling tools. They are analytically useful, but their outputs should not be treated as literal real-world blocs or definitive measures of geopolitical importance. They indicate structural patterns that require interpretation.

Finally, the report does not attempt formal causal inference or prediction. It identifies structural change, bridge roles, and cross-theater organization, but it does not claim to prove causal chains, hidden coordination, or future escalation.

19 Conclusion

This project studied the US–Israel–Iran conflict system as an evolving network rather than as a collection of isolated incidents. Through data harmonization, strict conflict-scope filtering, phase-wise segmentation, actor-network construction, community detection, and bridge analysis, the report developed a structural account of how the conflict system changes over time.

The main conclusion is that the recent phase is not merely larger in terms of event volume. It is structurally broader, more differentiated, and more multi-theater in character. More actors participate, more communities appear, and bridge positions become more important for maintaining connectivity across the wider system.

A second major conclusion is that structurally important actors are not always the same as the most active actors. This distinction is one of the most valuable outcomes of the project because it shows how network-science methods can reveal forms of geopolitical importance that simple descriptive counting tends to miss.

Overall, the contribution of the report lies in showing that standard network-science tools can generate meaningful insight into an ongoing geopolitical conflict system. The project remains methodologically manageable, but it still produces a non-trivial interpretation of escalation, fragmentation, and cross-theater connectivity in the contemporary US–Israel–Iran conflict system.

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